APPENDIX A

PROCEDURE FCR SIDE-WALL CORRECTICN

Introduction

If the wall surface is not as rough as the bed, as was the case in all the experimental runs with a sand bed, then the distribution of shear is nonuniform. As the bed becomes rougher, the shear on the bed increases and the shear on the wall decreases relative to the bed shear, because the bed offers more resistance to flow. In the central region the flow becomes more nearly two-dimensional, and the effect of the walls diminishes. Because of the variable effect of the walls caused by the changing bed roughness, a wall correction is needed to unify the results.

Several methods of side-wall correction are available. Einstein (41) has given a method based on the Manning flow equation, and Johnson (19) has presented one based on the Darcy-Weisbach formula. The Darcy-Weisbach friction factor has a sounder basis in modern fluid mechanics, and takes account of temperature. Johnson's method has been followed here, with some modifications to facilitate solution of the equations. An explanation of this method, as used, is presented below with an example.

Derivation of Equations for Side-Wall Correction

First, it will be convenient to define here a few new terms and to recapitulate some definitions and formulas already given.

S = slope of the energy line
ν = kinematic viscosity
g = acceleration of gravity
U = mean velocity

These quantities apply to either the entire flow, the bed section, or the wall section.

Each of the following quantities and formulas may be used either with subscript b for the bed section or subscript w for the wall section, or without any subscript for the whole channel:

A = area of cross section
p = wetted perimeter

$$r = A/p = \text{hydraulic radius} \qquad \text{(A-1)}$$

$$U_* = \sqrt{grS} = \text{shear or friction velocity} \qquad \text{(A-2)}$$

$$f = 8(U_*/U)^2 = \text{Darcy-Weisbach friction factor} \qquad \text{(A-3)}$$

$$R = 4\ Ur/\nu = \text{Reynolds number} \qquad \text{(A-4)}$$

The following assumptions are made:

1. The cross section can be divided into two sections, one producing shear on the bed and the other shear on the walls. The boundaries between the bed and wall sections are considered surfaces of zero shear, and are not included in the wetted perimeters p_b and p_w.

2. The velocity in the wall section, U_w, equals the velocity in the bed section, U_b.

3. The formulas above for r, U_*, f, and R can be applied to each section as if it were a channel by itself.

4. The roughnesses of the walls and bed are each homogeneous, although different.

The derivation which follows applies to any reasonable shape of open channel section. It may also be extended to cases with more than two types of roughness.

The following fundamental quantities are taken as known variables: U, S, g, ν, p_w, p_b and A. In addition, the quantities r, U_*, f, and R, for the whole section, may be determined directly by Eqs. A-1 to A-4. The principal unknowns of interest are U_{*b}, the friction velocity for the bed, r_b, the bed hydraulic radius, and f_b, the bed friction factor.

To solve the problem, f_w must also be known. If the equivalent sand roughness of the side walls is known, then the value of f_w can be determined with some auxiliary calculations from a generalized pipe resistance or Stanton diagram, such as the one given by Rouse (42). Since the side walls were practically hydrodynamically smooth in the present investigations, the procedure for finding f_w will be somewhat simplified by the elimination of the relative roughness as one of the variables. However, the general case can be solved with the same type of approach.

For smooth walls, f_w will be a function only of the Reynolds number for the wall which is:

$$R_w = \frac{4\ U_w r_w}{\nu} = \frac{4U\ r_w}{\nu}$$

Inasmuch as r_w is still unknown, rewrite R_w as

$$R_w = R\,\frac{r_w}{r} \tag{A-5}$$

By Eqs. A-2 and A-3,

$$g\ r_w\ S = \frac{f_w}{8}\ U^2 \tag{A-6}$$

and

$$g\ r\ S = \frac{f}{8}\ U^2\ . \tag{A-7}$$

Hence

$$\frac{r_w}{r} = \frac{f_w}{f}\ , \tag{A-8}$$

and Eq. A-5 becomes

$$\frac{R_w}{f_w} = \frac{R}{f}\ . \tag{A-9}$$

Thus the quantity $\frac{R_w}{f_w}$ can be found directly, but neither R_w nor f_w individually. Consequently, it was necessary to construct a special curve, Fig. 39, giving f as a function of R/f for smooth-walled channels. Fig. 39 is based on the graph of f vs. R given by Rouse for the Karman-Prandtl resistance equation for turbulent flow in smooth pipes. Hence, knowing f and R, it was possible to find f_w by Eq. A-9 and Fig. 39.

Now the geometry requires that

$$A = A_b + A_w\ . \tag{A-10}$$

If $r = A/p$ is substituted into Eq. A-7,

$$A = \frac{p\ f\ U^2}{8\ gS}\ .$$

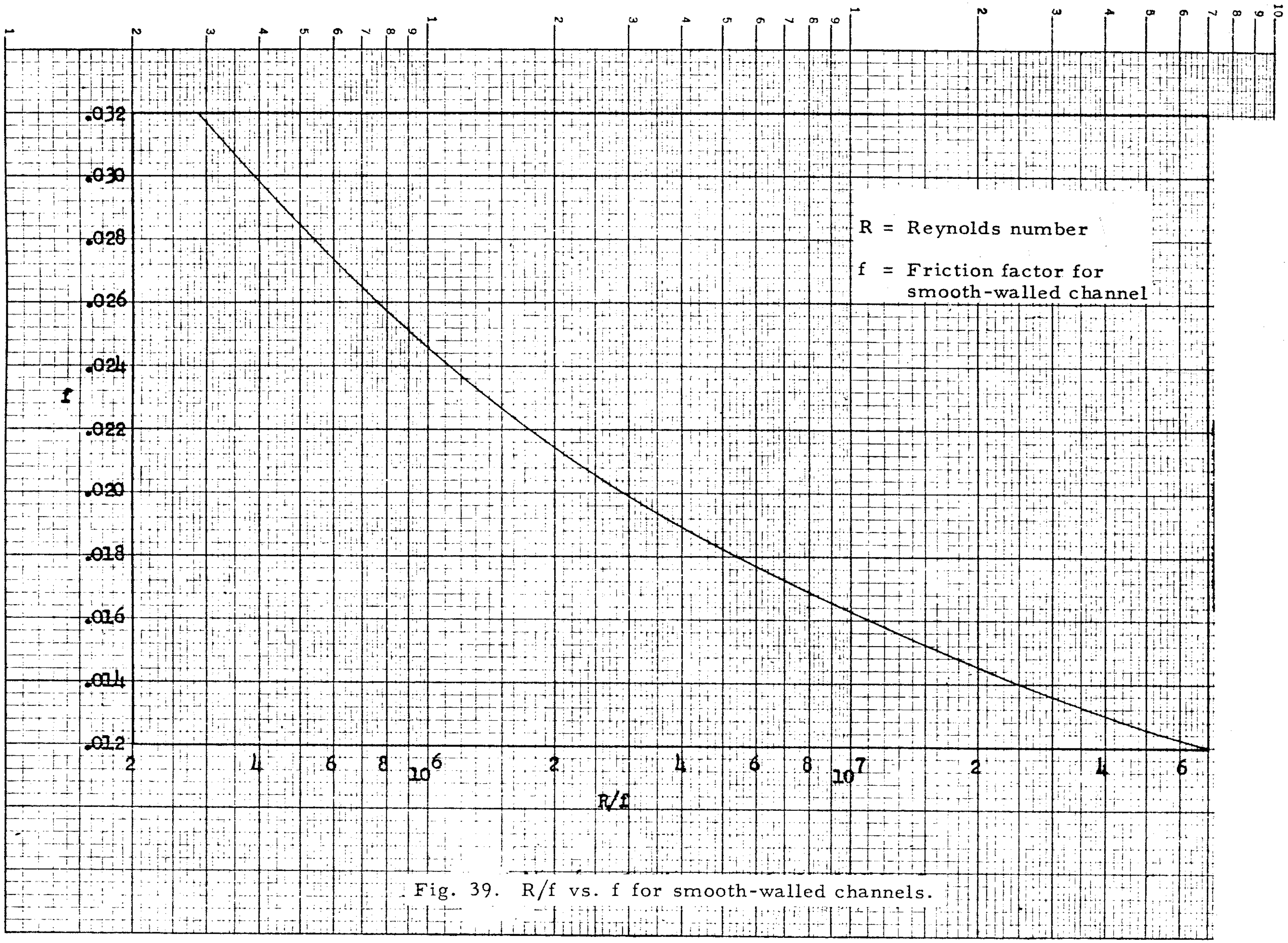

Fig. 39. R/f vs. f for smooth-walled channels.

Similarly, $$A_b = \frac{p_b f_b U^2}{8 g S}$$

and $$A_w = \frac{p_s f_w U^2}{8 g S} .$$

Putting these expressions for the area into Eq. A-10 and cancelling the common factor $U^2/2gS$, the result is

$$pf = p_b f_b + p_w f_w \qquad \text{(A-11)}$$

Since f_b is the only unknown, the solution is

$$f_b = \frac{p}{p_b} f - \frac{p_w}{p_b} f_w . \qquad \text{(A-12)}$$

For rectangular channels of width b and depth d, where

$$p = b + 2 d,$$

$$p_b = b,$$

and $$p_w = 2d,$$

Eq. A-12 may be further similified to

$$f_b = f + \frac{2d}{b}(f - f_w) \qquad \text{(A-13)}$$

By analogy with Eq. A-8,

$$r_b = r \frac{f_b}{f} . \qquad \text{(A-14)}$$

To complete the solution, U_{*b} may be found from its definition, Eq. A-2:

$$U_{*b} = \sqrt{g r_b S} . \qquad \text{(A-15)}$$

Any other desired quantities, such as r_w, A_w, and A_b may be readily found from the fundamental equations.

Sample Calculation

The application of this side-wall correction method will be illustrated for Run 2-1, for which the bed is covered with dunes. The pertinent quantities which were measured directly, by methods outlined in Chap. IV, are as follows:

$$Q = 0.855 \text{ cfs}$$
$$d = 0.240 \text{ ft}$$
$$b = 2.79 \text{ ft}$$
$$S = 0.00278$$
$$T = 25.5^\circ C$$

From these, the following quantities are calculated using Eqs. A-1 to A-4:

$$A = bd = (2.79)(0.240) = 0.670 \text{ sq ft}$$

$$U = Q/A = \frac{0.855}{0.670} = 1.28 \text{ ft/sec}$$

$$p = b + 2d = 2.79 + 2(0.240) = 3.27 \text{ ft}$$

$$r = A/p = \frac{0.67}{3.27} = 0.205 \text{ ft}$$

$$U_* = \sqrt{grS} = \sqrt{(32.2)(0.205)(.00278)} = 0.135 \text{ ft/sec}$$

$$f = 8\left(\frac{U_*}{U}\right)^2 = 8\,\frac{(0.135)^2}{(1.28)^2} = 0.090$$

$$\nu = 0.955 \times 10^{-5} \text{ ft}^2/\text{sec}$$

$$R = \frac{4\,U\,r}{\nu} = \frac{4(1.28)(.205)}{0.955 \times 10^{-5}} = 110{,}000\ .$$

The next step is to find f_w, the friction factor for the smooth walls. By Eq. A-9

$$\frac{R_w}{f_w} = \frac{R}{f} = \frac{110{,}000}{0.090} = 1{,}220{,}000\ .$$

From Fig. 39, the value $f_w = 0.0236$ is read from the curve for $R_w/f_w = 1.22 \times 10^6$.

Now by Eq. A-13

$$f_b = f + \frac{2d}{b}(f - f_w) = 0.090 + \frac{2(0.240)}{2.79}(0.090 - 0.0236)$$

$$f_b = 0.101\ ,$$

and by Eqs. A-14 and A-15,

$$r_b = r\,\frac{f_b}{f} = 0.205\,\frac{0.101}{0.090} = 0.231 \text{ ft}$$

$$U_{*b} = \sqrt{g r_b S} = \sqrt{(32.2)(0.231)(0.00278)} = 0.144 \text{ ft/sec}$$

The ratio of the mean bed shear stress to the over-all average is

$$\frac{\overline{\tau}_b}{\overline{\tau}_o} = \frac{r_b}{r} = \frac{f_b}{f} = \frac{0.101}{0.090} = 1.12 \;.$$

Similarly for the wall

$$\frac{\overline{\tau}_w}{\overline{\tau}_o} = \frac{r_w}{r} = \frac{f_w}{f} = \frac{0.0236}{0.090} = 0.26 \;.$$

Since the average bed shear is now almost four times the wall shear, it is clear that a very significant redistribution of shear stresses has been caused by the increased roughness of the bed due to the dunes.

A side-wall correction is obviously indispensable; Johnson's method (19) as used here is the simplest and most reliable one. A slight improvement of the method, allowing for different bed and wall section velocities, has been outlined by Brooks (18) but has not been used in reducing the experimental results because the increased labor of computation is not worthwhile.